

DEV_IMPL_GUIDE - SUB-WF-ACTIVATE-01-FLOW-WIK

1. Purpose

This guide explains how the WIK activation BPMN is implemented and tested.

Use it with:

- DD_SUB-WF-ACTIVATE-01-FLOW-WIK-v1.0.md
- DD_SUB-WF-ACTIVATE-01-Subscription_Activation-v1.0.md
- DD_SUB-WF-ACTIVATE-01-DEV_IMPL_GUIDE-v1.0.md

This guide focuses on the BPMN process file sub-activate-wik.bpmn.

2. Runtime Actors

Runtime component	Role
Camunda	Runs process sub-activate-wik.
subscription-workflow-api	Starts process through framework.
Worker apps	Execute external task topics.
Message correlators	Send payment and fulfillment messages to Camunda.

3. Before BPMN Deployment

Confirm:

1. subscription_operation_config has WIK ACTIVATE -> sub-activate-wik.
2. subscription_activation_config has WIK row.
3. Drools package rules.subscription.activate.wik is deployed.
4. Worker apps subscribe to required topics.
5. Message correlators can send payment-confirmed, fulfillment-completed, fulfillment-failed.

4. Step 1 - Create BPMN Process

Create:

```
sub-activate-wik.bpmn
```

Process key:

```
sub-activate-wik
```

Start event must expect variables from the framework:

```
{
  "operationId": "subop_01HZ_KAMAU_ACT",
  "operatorCode": "WIK",
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "operationKind": "ACTIVATE",
  "activationTrigger": "SALES_FIELD",
  "resolvedActivatedAt": "2026-05-18T10:00:00Z",
  "resolvedCycleAnchorDay": 18,
  "equipmentBindings": [
    {
      "equipmentRequirement": "GPON_ONT",
      "equipmentRef": "ont_01HZ_KAMAU"
    }
  ]
}
```

5. Step 2 - Add Happy Path Service Tasks

Add service tasks in this order:

Order	BPMN element	Camunda topic/message
1	Service task	drools-decide
2	Exclusive gateway	eligible?
3	Service task	sub-lm-put-pending-status
4	Service task	sub-lm-compute-cycle-window
5	Service task	bil01-charge-pre-authorize
6	Intermediate catch	payment-confirmed
7	Service task	bil01-confirm-charge
8	Service task	ful-call-activate
9	Intermediate catch	fulfillment-completed
10	Service task	ful-call-claim-homepass
11	Service task	osr-equipment-bind
12	Service task	sub-lm-put-master-fields
13	Service task	sub-lm-commit-state
14	Service task	kafka-emit
15	Service task	notification-send
16	Service task	subscription-operation-finalize
17	End event	success

6. Step 3 - Add Payment Timer

Attach boundary timer to payment-confirmed catch:

```
P7D
```

Timer route:

```
payment timeout
-> bil01-revert-pending-charge
-> sub-lm-revert-to-prior
-> kafka-emit SubscriptionActivationFailed
-> subscription-operation-finalize FAILED
```

7. Step 4 - Add Fulfillment Failure Route

Add catch for:

```
fulfillment-failed
```

Route:

```
fulfillment failed
-> bil01-revert-pending-charge
-> sub-lm-revert-to-prior
-> kafka-emit SubscriptionActivationFailed
-> subscription-operation-finalize FAILED
```

8. Step 5 - Add Equipment Bind Failure Route

If osr-equipment-bind fails after network activation:

```
equipment bind failed
-> fulfillment rollback topic if available
-> bil01-revert-pending-charge
-> sub-lm-revert-to-prior
-> kafka-emit SubscriptionActivationFailed
-> subscription-operation-finalize FAILED
```

If fulfillment rollback is not available, route to manual recovery.

9. Step 6 - Add Manual Recovery

For repeated sub-lm-commit-state failure, open:

```
manual-recovery-decision
```

Do not mark the operation COMPLETED until a human resolves the final state.

10. Step 7 - Deploy BPMN

Deploy through CI/CD or Camunda deployment API:

```
POST /engine-rest/deployment/create
Content-Type: multipart/form-data
```

Verify:

```
GET /engine-rest/process-definition/key/sub-activate-wik
```

Expected:

```
{
  "key": "sub-activate-wik",
  "suspended": false
}
```

11. Step 8 - Test Happy Path

1. Start activation through API.
2. Confirm process instance exists.
3. Let workers process to payment wait.
4. Correlate payment-confirmed.
5. Let billing confirm charge.
6. Let FUL worker submit activation.
7. Correlate fulfillment-completed.
8. Let workers complete HomePass, equipment, SUB-LM commit, event, notification, finalize.
9. Confirm operation is COMPLETED.
10. Confirm subscription is ACTIVE.

12. Step 9 - Test Failure Paths

Test:

- Drools rejection.
- Payment timeout.
- Fulfillment failed.
- Equipment bind failed.
- Notification failed.
- SUB-LM commit failed.

Expected:

- Failure paths finalize operation as FAILED, except manual recovery path.
- Subscription returns to CREATED where compensation applies.
- Success event is not emitted on failure.

13. Common Mistakes

- Forgetting the payment boundary timer.
- Emitting SubscriptionActivated before sub-lm-commit-state.
- Treating notification-send failure as fatal.
- Missing fulfillment-failed catch.
- Missing compensation event SubscriptionActivationFailed.

- Using topic names as deployment names.